

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1105

Pacific Way 138 kV
January 28, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1105. The Interconnection Customer proposes a 541.3 MW natural gas combined cycle facility interconnecting at the Pacific Way 138 kV substation in the Pennsylvania area. The request was received on November 17, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 417 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$40,599,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Wec 3 - Crystal Falls 138kV
Contingency: Ritter Pak - Sugar Hill 345kV line
Loading: 305.7% (Rating: 168 MVA)
DFAX: 59.3%
Pre-project loading: 114.9%
2. Facility: Crystal Falls - Pacific Way 138kV
Contingency: Sugar Hill - Ritter Park 345kV L
Loading: 252.5% (Rating: 213 MVA)
DFAX: 59.3%
Pre-project loading: 101.7%
3. Facility: Bollman Tap 2 - Pacific Way 138kV
Contingency: Crystal Falls - Pacific Way Substation 138kV
Loading: 232.3% (Rating: 227 MVA)
DFAX: 100.0%
Pre-project loading: 5.6%
4. Facility: Wec 3 - Westridge Windfarm Llc 138kV
Contingency: Ritter Park - Sugar Hill 345kV
Loading: 231.8% (Rating: 211 MVA)
DFAX: 59.3%
Pre-project loading: 79.8%
5. Facility: Ontonagon - Roseland 138kV
Contingency: Piney Crk Project - Hilltop 500kV
Loading: 227.2% (Rating: 179 MVA)
DFAX: 5.2%
Pre-project loading: 211.7%

6. Facility: Oceano Substation - Bollman Ta 2 138kV Line
Contingency: Pacific Way - Crystal Falls 138kV line
Loading: 224.9% (Rating: 232 MVA)
DFAX: 100.0%
Pre-project loading: 8.3%
7. Facility: Mon Valley Works Illinois - Roseland 138kV Line
Contingency: Hilltop - Piney Crk Project 500kV
Loading: 201.4% (Rating: 198 MVA)
DFAX: 5.2%
Pre-project loading: 187.3%
8. Facility: Westridge Windfarm Llc - Ramon 138kV
Contingency: Ritter Park - Sugar Hill 345kV
Loading: 199.9% (Rating: 234 MVA)
DFAX: 59.3%
Pre-project loading: 62.6%
9. Facility: North Grafton - Oceano 138kV Line
Contingency: Pacific Way - Crystal Falls 138kV
Loading: 196.4% (Rating: 253 MVA)
DFAX: 100.0%
Pre-project loading: 17.3%
10. Facility: Brunot Island - Ramon 138kV
Contingency: Sugar Hill - Ritter Park 345kV
Loading: 191.9% (Rating: 240 MVA)
DFAX: 59.3%
Pre-project loading: 58.2%
11. Facility: North Grafton - Ulmerton Transfer Center 138kV
Contingency: Crystal Falls - Pacific Way 138kV
Loading: 190.3% (Rating: 257 MVA)
DFAX: 100.0%
Pre-project loading: 19.9%
12. Facility: V C Summer - Mon Vly Works Illinois 138kV Line
Contingency: Hilltop - Piney Crk Project Substation 500kV
Loading: 175.4% (Rating: 218 MVA)
DFAX: 5.2%
Pre-project loading: 162.6%
13. Facility: Anson County Generation Facility - Us Gypsum 138kV line
Contingency: Leicester Jct - Woodville 138kV
Loading: 155.4% (Rating: 218 MVA)
DFAX: 10.8%
Pre-project loading: 128.5%
14. Facility: Oklahoma Gas Electric - Kyles Ford 138kV
Contingency: Conrath Dpc Solar - Brunot Island Substation 138kV
Loading: 155.0% (Rating: 177 MVA)
DFAX: 44.7%
Pre-project loading: 18.0%
15. Facility: Anson County Generation Facility - Walmart Substation 138kV line
Contingency: Leicester Jct - Woodville 138kV
Loading: 152.4% (Rating: 233 MVA)
DFAX: 10.8%
Pre-project loading: 127.3%
16. Facility: Oklahoma Gas Electric - Sioux Falls 138kV
Contingency: Brunot Island - Conrath Dpc Solar 138kV

- Loading: 149.6% (Rating: 191 MVA)
DFAX: 44.7%
Pre-project loading: 23.3%
17. Facility: Brunot Island - Conrath Dpc Solar 138kV
Contingency: Brunot Island - Sioux Falls 138kV line
Loading: 147.1% (Rating: 264 MVA)
DFAX: 58.4%
Pre-project loading: 27.2%
18. Facility: Spring Ridge Rd - V C Summer 138kV
Contingency: Hilltop - Piney Creek Project 500kV Line
Loading: 143.8% (Rating: 246 MVA)
DFAX: 5.2%
Pre-project loading: 132.5%
19. Facility: Dos Amigos - Ontonagon 138kV L
Contingency: Conrath Dpc Solar - Sierra Brooks 138kV line
Loading: 143.2% (Rating: 239 MVA)
DFAX: 9.1%
Pre-project loading: 122.6%
20. Facility: Arsenal Substation - East Kent 138kV
Contingency: Us Gypsum - Forbes 138kV
Loading: 141.9% (Rating: 224 MVA)
DFAX: 5.9%
Pre-project loading: 127.7%
21. Facility: Sioux Falls - Brunot Island 138kV
Contingency: Brunot Island - Conrath Dpc Solar 138kV
Loading: 140.8% (Rating: 215 MVA)
DFAX: 44.7%
Pre-project loading: 28.4%
22. Facility: Arsenal - Wyandotte 138kV
Contingency: Us Gypsum - Forbes 138kV
Loading: 139.7% (Rating: 239 MVA)
DFAX: 5.9%
Pre-project loading: 126.5%
23. Facility: E Kent - Saline 138kV L
Contingency: Us Gypsum - Forbes 138kV
Loading: 139.7% (Rating: 195 MVA)
DFAX: 5.9%
Pre-project loading: 123.5%
24. Facility: Williams - Wyandotte 138kV L
Contingency: Us Gypsum - Forbes Substation 138kV Line
Loading: 139.3% (Rating: 246 MVA)
DFAX: 5.9%
Pre-project loading: 126.4%
25. Facility: Spring Ridge Road - E Omak Substation 138kV
Contingency: Piney Creek Project Substation - Hilltop 500kV L
Loading: 138.9% (Rating: 250 MVA)
DFAX: 5.2%
Pre-project loading: 127.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wec 3 - Crystal Falls 138kV	Rebuild line with higher-capacity conductor	\$714,000	100% (\$714,000)
Crystal Falls - Pacific Way 138kV	Rebuild line with higher-capacity conductor	\$1,279,000	100% (\$1,279,000)
Bollman Tap 2 - Pacific Way 138kV	Rebuild line with higher-capacity conductor	\$2,974,000	100% (\$2,974,000)
Wec 3 - Westridge Windfarm Llc 138kV	Rebuild line with higher-capacity conductor	\$465,000	100% (\$465,000)
Roseland - Ontonagon 138kV	Rebuild line with higher-capacity conductor	\$1,109,000	70% (\$780,000)
Oceano Substation - Bollman Ta 2 138kV Line	Rebuild line with higher-capacity conductor	\$1,622,000	100% (\$1,622,000)
Mon Valley Works Illinois - Roseland 138kV Line	Rebuild line with higher-capacity conductor	\$2,938,000	70% (\$2,068,000)
Westridge Windfarm Llc - Ramon 138kV Line	Rebuild line with higher-capacity conductor	\$1,354,000	100% (\$1,354,000)
North Grafton - Oceano 138kV Line	Rebuild line with higher-capacity conductor	\$2,251,000	100% (\$2,251,000)
Brunot Island - Ramon 138kV	Rebuild line with higher-capacity conductor	\$3,664,000	100% (\$3,664,000)
N Grafton - Ulmerton Transfe Center 138kV	Rebuild line with higher-capacity conductor	\$2,806,000	100% (\$2,806,000)
Mon Vly Works Illinois - V C Summer 138kV	Rebuild line with higher-capacity conductor	\$2,322,000	70% (\$1,634,000)
Anson County Generation Facility - Us Gypsum 138kV line	Rebuild line with higher-capacity conductor	\$969,000	100% (\$969,000)
Oklahoma Gas Electric - Kyles Ford 138kV	Rebuild line with higher-capacity conductor	\$1,349,000	100% (\$1,349,000)
Anson County Generation Facility - Walmart Substation 138kV line	Rebuild line with higher-capacity conductor	\$629,000	100% (\$629,000)
Sioux Falls - Oklahoma Gas Electric 138kV	Rebuild line with higher-capacity conductor	\$1,351,000	100% (\$1,351,000)
Brunot Island - Conrath Dpc Solar 138kV	Rebuild line with higher-capacity conductor	\$3,458,000	100% (\$3,458,000)
Spring Ridge Rd - V C Summer 138kV Line	Rebuild line with higher-capacity conductor	\$5,000,000	70% (\$3,518,000)
Dos Amigos - Ontonagon 138kV L	Rebuild line with higher-capacity conductor	\$977,000	69% (\$676,000)
Arsenal - East Kent 138kV	Rebuild line with higher-capacity conductor	\$3,050,000	71% (\$2,177,000)
Sioux Falls - Brunot Island 138kV line	Rebuild line with higher-capacity conductor	\$1,305,000	100% (\$1,305,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Arsenal - Wyandotte 138kV	Rebuild line with higher-capacity conductor	\$922,000	71% (\$658,000)
E Kent - Saline 138kV L	Rebuild line with higher-capacity conductor	\$1,311,000	71% (\$936,000)
Williams - Wyandotte 138kV L	Rebuild line with higher-capacity conductor	\$802,000	71% (\$573,000)
Spring Ridge Road - East Omak 138kV	Rebuild line with higher-capacity conductor	\$1,974,000	70% (\$1,389,000)
	Total allocated to project		\$40,599,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.